Week #9: [Leg]

(Hyperlinked = anatomy notes below)

• Introduction to Module

LEg

- **Introduction to the Leg (18 min)**
- **Muscles (27 min)**
- **Neurovasculature (20 min)**



Anatomy of the Leg

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Schulich MEDICINE & DENTISTRY

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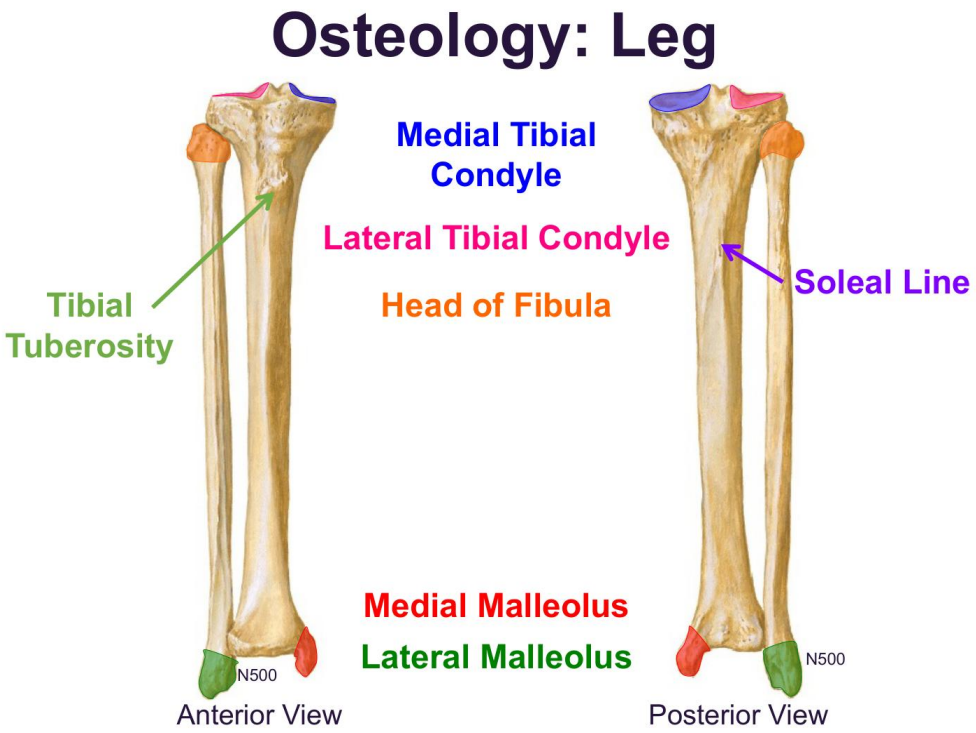
Intro to the Leg

- Distal Lower Extremity Osteology
- Organization of the Leg
- Movements at the ankle and foot

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Introduction to the Leg

- Distal Lower Extremity Osteology
- Organization of the Leg
- Movements at the Ankle and Foot



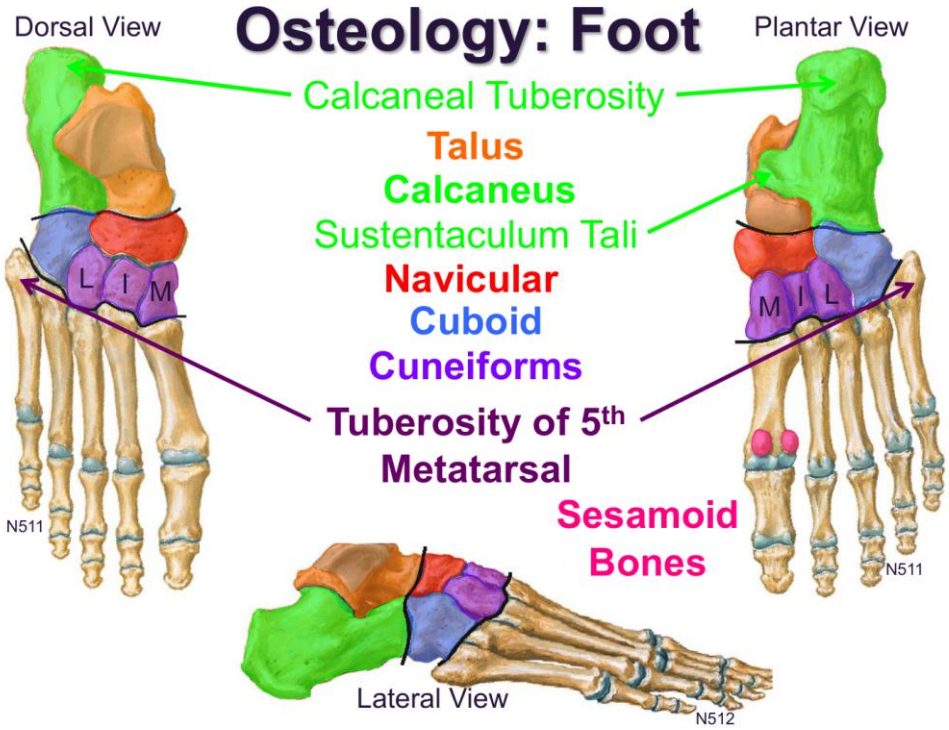
Clinical Note: **Leg Fractures** - The tibia is most susceptible to fracture at its narrowest point between the middle and distal thirds, which is also the area with the poorest blood supply. The fibula is most likely to fracture just proximal to the lateral malleolus due to pushing of the talus during extreme inversion of the foot. Also, because of the strong interosseous membrane that connects the fibula and tibia, when the tibia breaks the fibula will sometimes break with it.

I. Osteology of the Leg

A. Tibia – The Tibia is the major weight bearing bone of the leg. It extends superficially into the subcutaneous tissue in the anteromedial leg earning its nickname the "shin bone." Because of its superficial location, the periosteum of the tibia is very susceptible to bruising.

a. **Medial Condyle and Lateral Condyle** –make a wide articulation with the femur in order to provide for proper weight transfer. The ***tibial plateau*** is formed by the articular surfaces of the two condyles.
b. **Tibial Tuberosity** – Anterior protrusion that is the insertion site of the patellar ligament.
c. **Shaft** –provides a place of origin for a number of leg muscles. The ***soleal line*** lies on the posterior side of the shaft and is the sight of origin of the soleus muscle.
d. **Medial Malleolus** – widening of the tibia distally for articulates with and is involved in weight transfer to the talus.

Fibula – The fibula lies posterior and lateral to the tibia. The **head of the fibula** articulates with the tibia proximally and the **lateral malleolus** articulates with the talus distally to help hold the talus in place with the medial malleolus of the tibia. The fibula does not have a weight bearing function, and for this reason, is the bone of choice in bone grafts (removal of part of the fibula does not necessarily take away any ability to walk, run or jump)



OSTEOLOGY

A. Seven Tarsals:

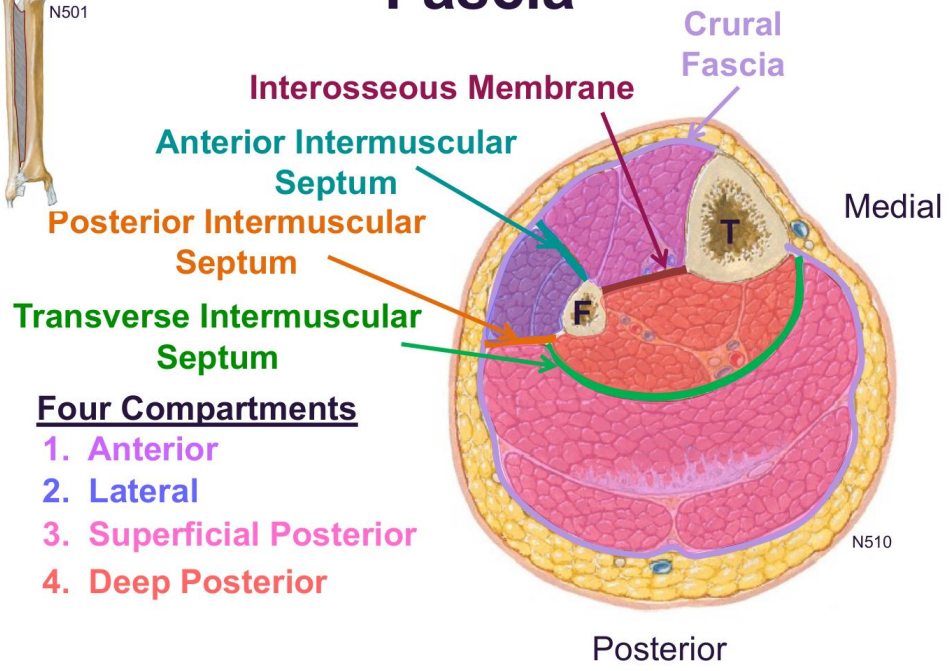
1. **Talus** – The talus is held in place by the medial and lateral malleoli of the tibia and fibula respectively. It articulates with the calcaneus inferiorly and the navicular anteriorly. Its function is to transfer weight from the tibia to both the calcaneus and forefoot via articulation and the forefoot via ligaments.
2. **Calcaneus** – The calcaneus forms the heel and is the largest bone of the foot. It transfers weight from the talus to the ground. It articulates with the cuboid and the talus. The **calcaneal tuberosity**, located on the posterior surface of the calcaneus, is the insertion site for the calcaneal tendon ("Achilles Tendon"); and the **sustentaculum tali**, located superomedially, provides a shelf for the talus to sit on as well as a pulley for the flexor hallucis longus tendon to pass under.
3. **Navicular** – A boat shaped tarsal that articulates with the talus posteriorly and the three cuneiform bones anteriorly. The navicular tuberosity, a prominence on the inferior aspect of the navicular, provides a key attachment site of the calcaneonavicular ligament ("Spring Ligament"), which helps to maintain the longitudinal arch along the medial foot.
4. **Cuboid** – the most lateral of the tarsal bones. The cuboid articulates with the calcaneus, talus, lateral cuneiform, navicular, and 4th and 5th metatarsals.
5. **Medial, Intermediate, and Lateral Cuneiform (First, Second and Third Cuneiform)** – Articulate with the navicular and cuboid posteriorly and the first four metatarsals anteriorly. Their wedge shape contributes greatly to the formation and maintenance of the transverse arch of the foot.
6. **Metatarsals and Phalanges** – Provide attachment sites for a number of muscles that move the foot and its digits. The **tuberosity of the 5th metatarsal** extends posteriorly off the 5th metatarsal and serves as an insertion site for the fibularis brevis. When the foot is inverted quickly, the fibularis brevis can pull the tuberosity of the 5th metatarsal off (avulsion fracture).

B. Sesamoid Bones – These bones sometimes form in tendons where the tendon rubs against bone. The patella is an example of a large sesamoid bone that formed in the tendon of the quadriceps as this tendon passes over the knee. In the foot, the tendons of flexor hallucis brevis will often form sesamoid bones as they pass over the 1st metatarsophalangeal joint.

a. **Clinical Note:** **Metatarsal fractures** occur when a heavy object falls on the foot. Metatarsal fractures are also common in dancers, a "dancer's fracture" can occur when a ballet dancer loses their balance and puts all their weight on their metatarsals causing a fracture of a metatarsal or multiple metatarsals.



Fascia



Four Compartments

1. Anterior
2. Lateral
3. Superficial Posterior
4. Deep Posterior

Leg Muscle Movements











III. Organization of the Leg: Fascia and Compartments

a. Deep fascia of the leg and foot is dense forming the **crural fascia** of the leg and **plantar aponeurosis** of the sole of the foot. Laminae of the deep fascia form septa that group muscles together into compartments of the leg and foot.

b. The leg is divided into an **anterior, posterior** (superficial and deep), and **lateral compartment** by the tibia, fibula, **interosseous membrane (IN), anterior intermuscular septum (AC) and posterior intermuscular septum (PC).** The crural fascia, a layer of deep fascia, forms a circle around the outside of the three compartments.

c. The posterior compartment is subdivided by the **transverse intermuscular septum** into the **superficial** and **deep posterior compartments.**

d. All tendons from the anterior and lateral compartments pass under a thickening of fascia, the **retinacula**, that bind the tendons before and after they cross the ankle joint preventing them from bowstringing anteriorly during dorsiflexion. **The are four retinacula:**

   i. Two bind tendons of the **anterior compartment:**

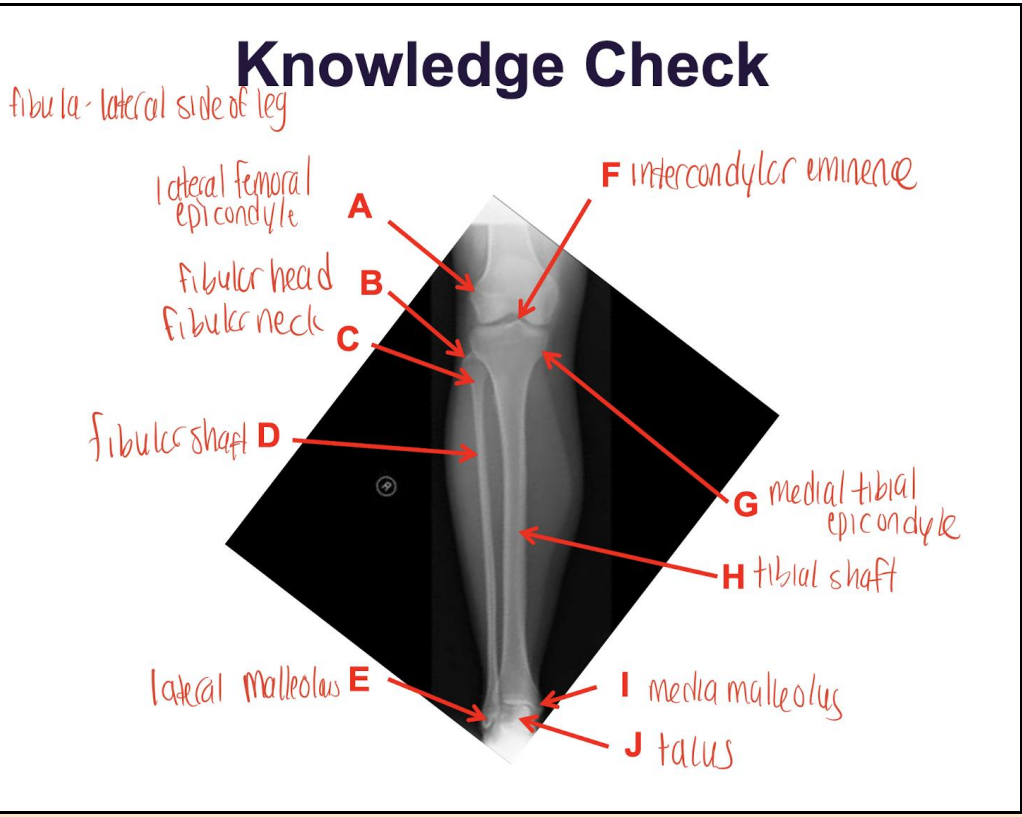
      1. **Superior Extensor Retinaculum** – travels from the tibia to the fibula anteriorly
      2. **Inferior Extensor Retinaculum** – Y-shaped with stem arising from calcaneus, the superior branch going to the medial malleolus, and the inferior branch going to deep fascia.

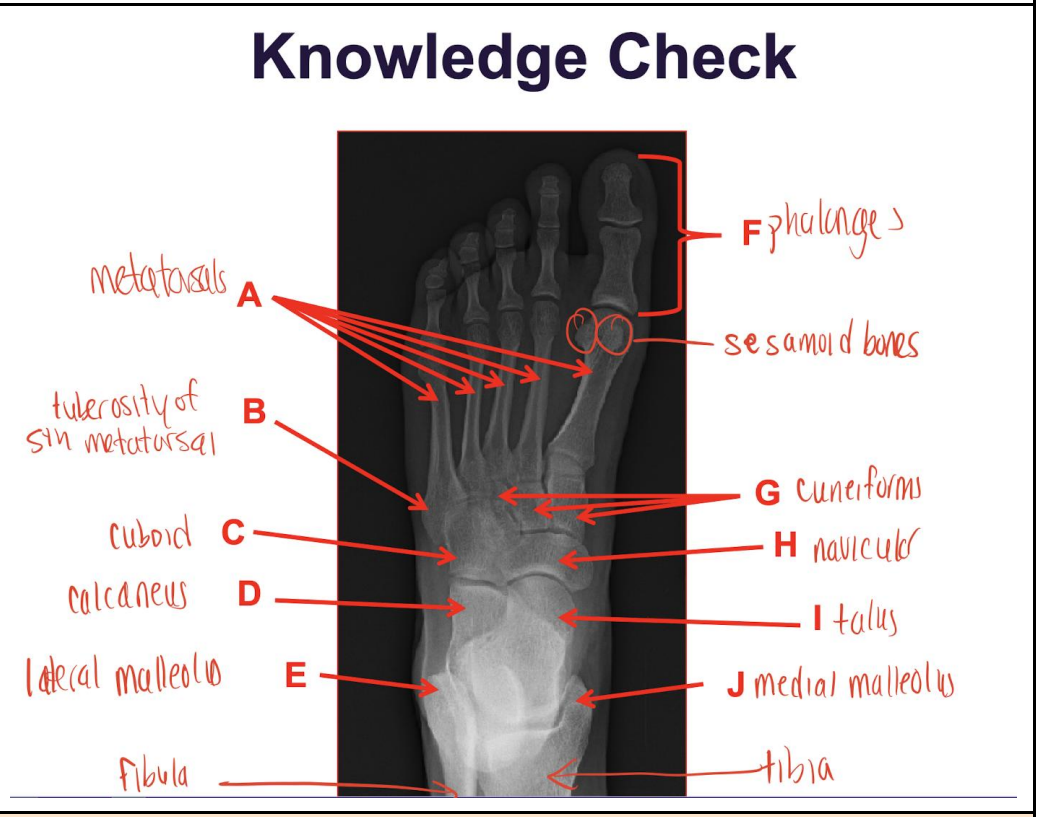
   ii. Two bind tendons of the **lateral compartment:**

      1. **Superior Fibular Retinaculum** – travels from the lateral malleolus to the calcaneus
      2. **Inferior Fibular Retinaculum** – travels from the calcaneus over the tendons of the Fibularis Longus and the Fibularis Brevis before inserting back on the calcaneus

Clinical Note: Compartment Syndrome/Containment and Spread of Compartmental Infections: The fascia in the leg seals the anterior, lateral, and posterior compartments at the joints not allowing the expansion of these compartments or the travel of infection between compartments. Pus from infection or inflammation from strenuous or unaccustomed exercise may increase the pressure within the compartment. This buildup of pressure may cause the nerves in that compartment to become pinched and the arteries compressed causing a painful condition in which there is little supply of nutrients and oxygen to the bones or muscle. A **fasciotomy** (incision of

fascia) may be necessary to relieve pressure and debride pockets of infection.





Muscles

- Anterior Compartment

Muscles: Anterior Compartment

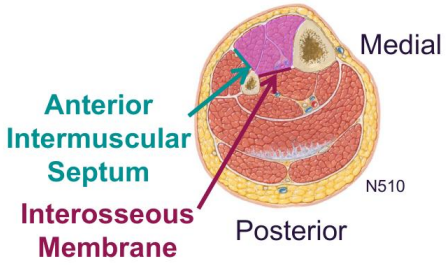


- Four Muscles
- Primarily responsible for **Dorsiflexion** of the ankle
- All innervated by: **Deep Fibular Nerve**

Extensor Retinacula

Superior Extensor Retinaculum

Inferior Extensor Retinaculum



Tibialis Anterior



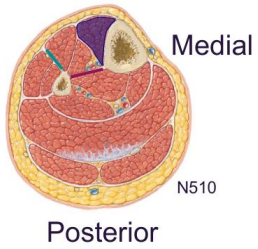
Myotome: L4 & L5



Origin:
Lateral Condyle, Proximal ½ of Lateral Tibia, & Interosseous Membrane

Insertion:
Plantar Aspect of Medial Cuneiform & Base of 1st MT

- Function:
  - Dorsiflexes Ankle
  - Inverts the Foot



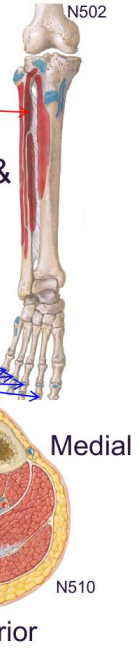
IV. Muscles

A. Anterior Compartment

a. There are four muscles in the anterior compartment, they are primarily responsible for **dorsiflexion** of the ankle, and they all have a common innervation: the **deep fibular nerve.**

Extensor Digitorum Longus



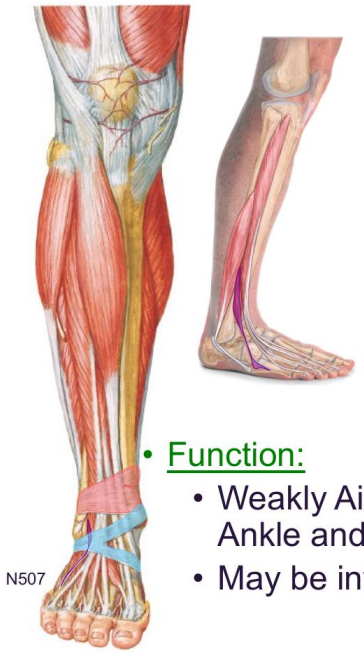


Origin:
Lateral Condyle of Tibia, Superior ¾ of Medial Fibula, & Interosseous Membrane

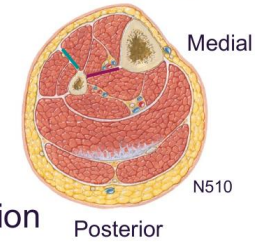
Insertion:
Middle & Distal Phalanges of Lateral Four Digits

- Function:
  - Dorsiflexes Ankle
  - Extends Lateral Four Digits

Peroneus (Fibularis) Tertius







Origin:
Inferior ⅓ of Anterior Fibula and Interosseous Membrane

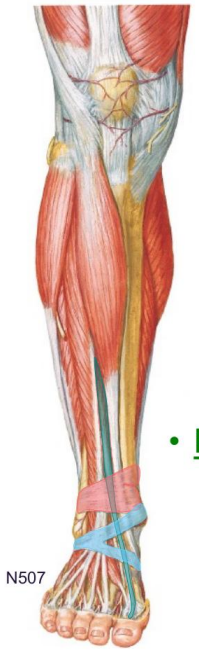
Insertion:
Dorsum of Base of 5th MT

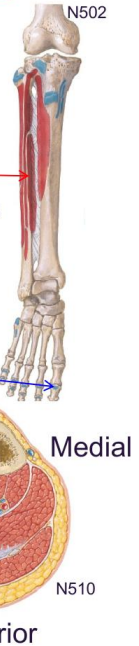
- Function:
  - Weakly Aids in Dorsiflexion of Ankle and Eversion of the Foot
  - May be involved in proprioception

*Not in cross-section

Extensor Hallucis Longus

Myotome: L5





Origin:
Middle ⅓ of Anterior Fibula, & Interosseous Membrane

Insertion:
Dorsal Aspect of Base of Distal Phalanx of 1st Digit (Great Toe)

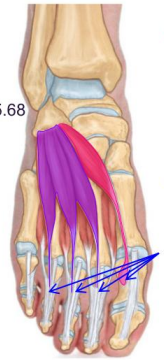
- Function:
  - Dorsiflexes Ankle
  - Extends 1st Digit (Great Toe)

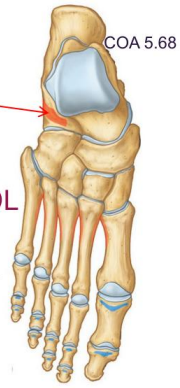
Clinical Note: **Shin Splints** - Microtrauma to the periosteum of the tibia where the muscles of the anterior compartment attach (especially Tibialis Anterior) causes pain in the anterior leg. Usually such trauma is caused by overexertion of the muscles. Shin splints are a mild form of anterior compartment syndrome. Inactive people are at risk for shin splints when they walk or run long distances; and runners are at risk when they improperly warm up or cool down.

Tibialis Anterior	Lateral Condyle, Proximal ½ of Lateral Tibia, and Interosseous Membrane	Plantar Aspect of Medial Cuneiform and Base of 1st Metatarsal	Deep Fibular Nerve	Dorsiflexes and Inverts the foot
Extensor Digitorum Longus	Lateral Condyle of Tibia, Superior, ¾ of Medial Fibula, and Interosseous Membrane	Middle and Distal Phalanges of digits 2-5	Deep Fibular Nerve	Dorsiflexes Ankle and Extends digits 2-5
Extensor Hallucis Longus	Middle ⅓ of Anterior Fibula and Interosseous Membrane	Dorsum of Base of Distal Phalanx of 1st Digit	Deep Fibular Nerve	Dorsiflexes Ankle and Extends 1st Digit
Fibularis Tertius*	Inferior ⅓ of Anterior Fibula and Interosseous Membrane	Dorsum of Base of 5th Metatarsal	Deep Fibular Nerve	Weakly aids in Dorsiflexion of Ankle and Foot Eversion May be involved in proprioception

Extensor Digitorum Brevis & Extensor Hallucis Brevis







Origin:
Calcaneus and stem of inferior extensor retinaculum

Insertion:
EDB: Extensor Expansion of EDL For Digits 2-4
EHB: Base of Proximal Phalanx of 1st Digit

- Function:
  - EDB: Extends Digits 2-4 with EDL
  - EHB: Extends 1st Digit with EHL
- Innervation:
  - Deep Fibular Nerve

Dorsum of the Foot

Muscle	Origin	Insertion	Innervation	Action
Extensor Digitorum Brevis	Calcaneus and Stem of Inferior Extensor Retinaculum	Tendons of Extensor Digitorum Longus Digits 2-4	Deep Fibular Nerve	Extends digits 2-4 with Extensor Digitorum Longus
Extensor Hallucis Brevis	Calcaneus	Proximal Phalanx of Big Toe	Deep Fibular Nerve	Extends 1st Digit with Extensor Hallucis Longus

Muscles

- Lateral Component

Muscles: Lateral Compartment

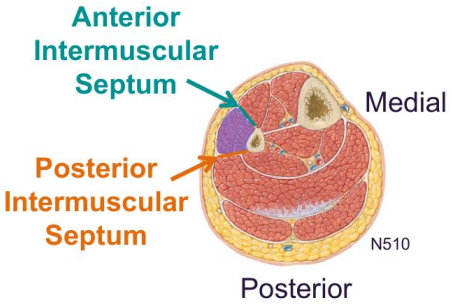


- Two Muscles
- Primarily responsible for **Eversion** of the ankle
- All innervated by: **Superficial Fibular Nerve**

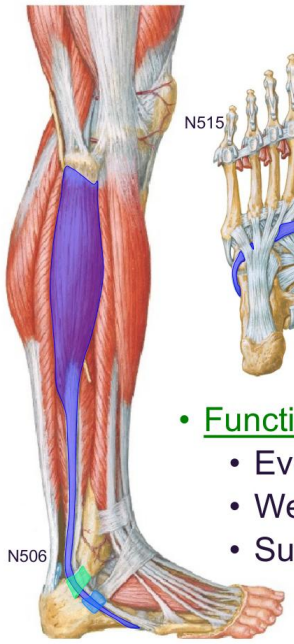
Fibular Retinacula

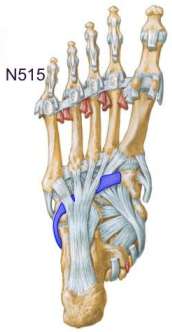
Superior Fibular Retinaculum

Inferior Fibular Retinaculum



Peroneus (Fibularis) Longus



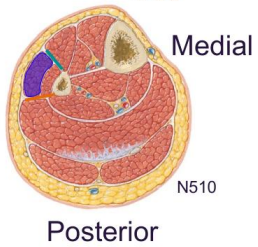


Origin:
Head and Superior ⅔ of Lateral Fibula

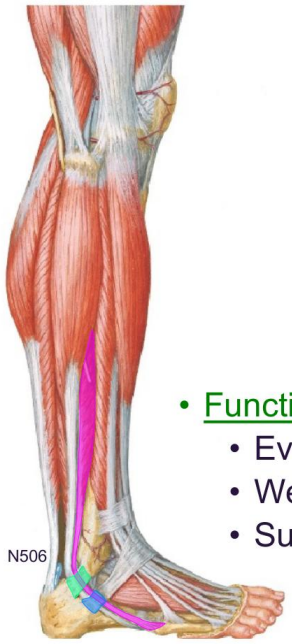
Insertion:
Medial Plantar Foot



- Function:
  - Everts Foot
  - Weakly Plantarflexes Ankle
  - Supports Transverse Arch



Peroneus (Fibularis) Brevis

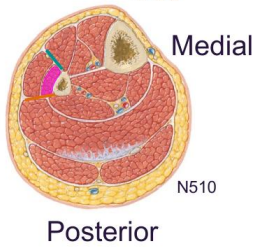


Origin:
Inferior ⅔ of Lateral Fibula

Insertion:
Base of 5th MT



- Function:
  - Everts Foot
  - Weakly Plantarflexes Ankle
  - Supports Transverse Arch





B. Lateral Compartment

a. There are two muscles in the lateral compartment, they are primarily responsible for **eversion** of the foot, and they all have a common innervation: the **superficial fibular nerve.**

Muscle	Origin	Insertion	Innervation	Action
Fibularis Longus	Head and Superior 2/3 of Lateral Fibula	Plantar Aspect of Base of 1st Metatarsal and Medial Cuneiform	Superficial Fibular Nerve	Everts Foot, Weakly Plantarflexes Ankle, and Supports Transverse Arch of Foot
Fibularis Brevis	Inferior 2/3 of Lateral Fibula	Dorsal surface of Tuberosity of 5th Metatarsal	Superficial Fibular Nerve	Everts Foot, Weakly Plantarflexes Ankle, and Supports Transverse Arch of Foot

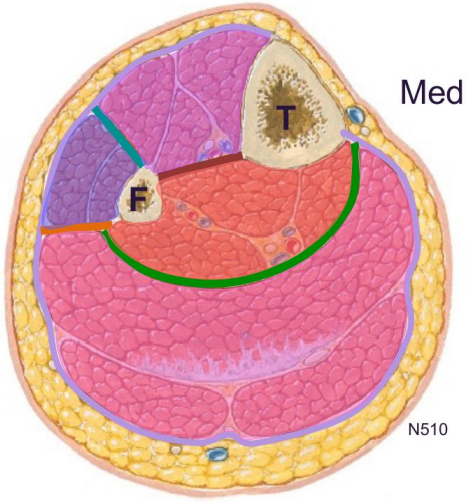
Muscles

- Posterior Compartment

Fascia

Four Compartments

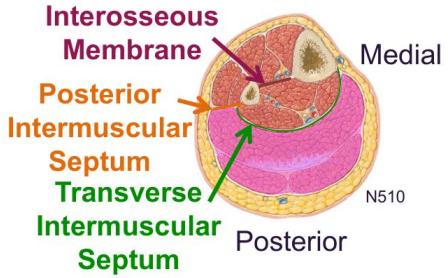
1. Anterior
2. Lateral
3. Superficial Posterior
4. Deep Posterior



Muscles: Superficial Posterior Compartment



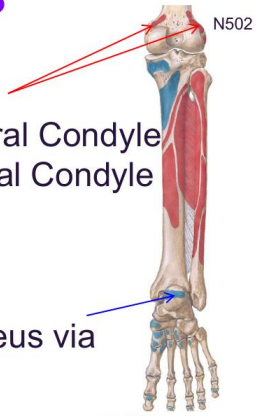
- Three Muscles
- Primarily responsible for **Plantarflexion** of the ankle
- All innervated by: **Tibial Nerve**



Gastrocnemius

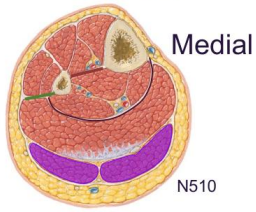


Myotome: S1



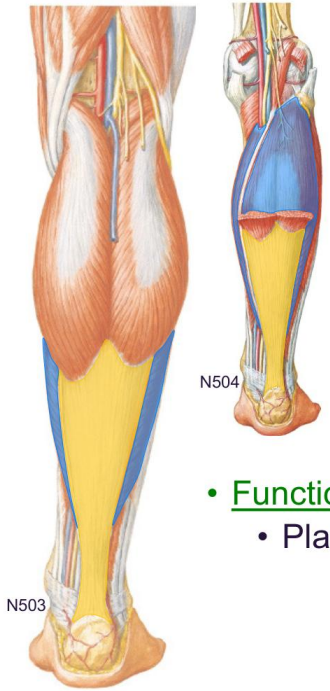
Origin:
Lateral Head: Lateral Femoral Condyle
Medial Head: Medial Femoral Condyle

Insertion:
Posterior Surface of Calcaneus via Calcaneal Tendon

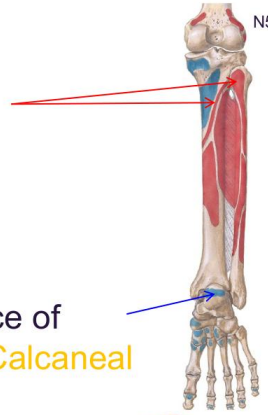


- Function:
  - Plantarflexes Ankle With Knee Extended
  - Raises Heel While Walking
  - Flexes Leg at Knee

Soleus

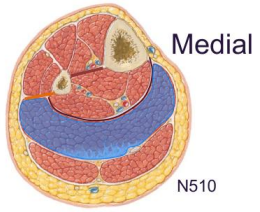


Myotome: S1

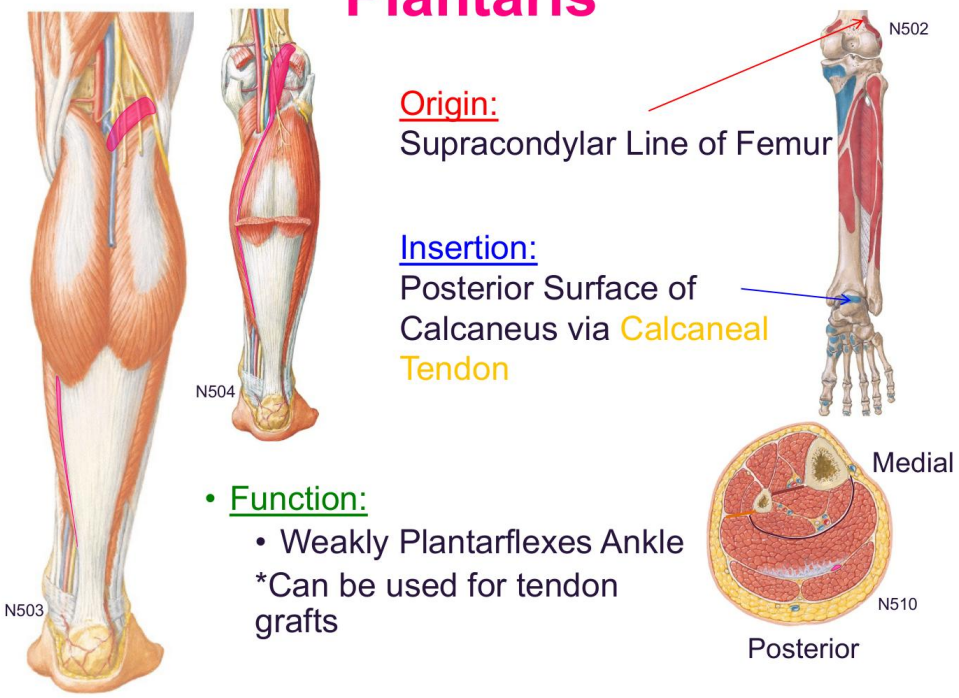


Origin:
Tibia & Fibula

Insertion:
Posterior Surface of Calcaneus via Calcaneal Tendon



- Function:
  - Plantarflexes Ankle



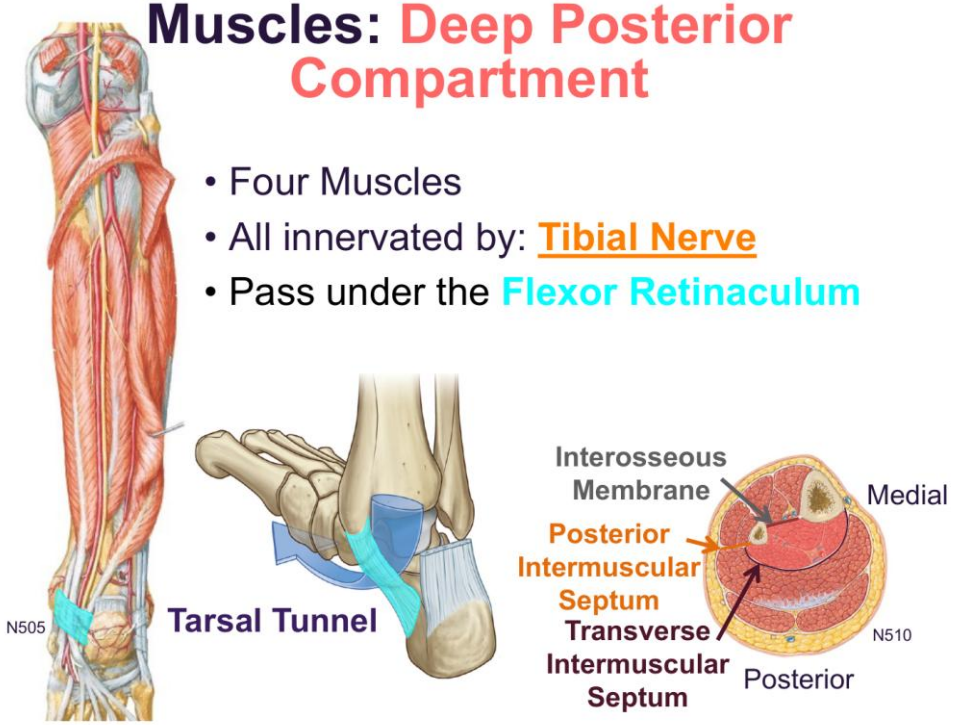
D. Superficial Posterior Compartment

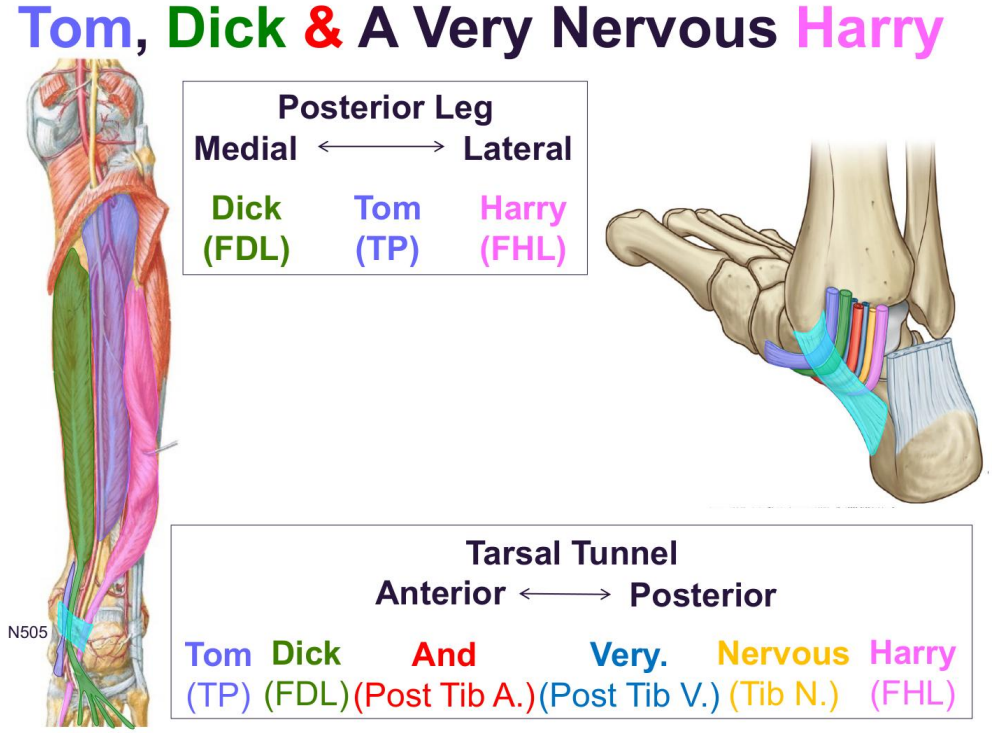
1. There are three muscles in the superficial posterior compartment, they are all primarily responsible for **plantarflexion**, and they all have a common innervation: the **tibial nerve.**

Clinical Note: Calcaneal Tendonitis - tendonitis can occur due to inflammation of the calcaneal tendon. This happens when there are microscopic tears in the tendon, especially a little superior to its attachment to the calcaneus. This is a common injury in runners. People who suffer long term tendonitis can experience calcaneal tendon rupture. If the tendon tears completely, the individual is unable to plantarflex against resistance.

Clinical Note: Calcaneal Tendon Rupture – usually occurs in poorly conditioned individuals. Mechanism of injury usually includes a forceful plantarflexion. The individual may experience an audible snap followed by sudden calf pain and dorsiflexion. In a completely ruptured tendon, a gap is palpable. Individuals with a calcaneal tendon rupture cannot plantarflex against resistance. In non-athletic/ older patients non surgical repairs may be adequate; however, in the younger more active population surgical intervention is usually advised.

Muscle	Origin	Insertion	Innervation	Action
Gastrocnemius	Lateral Head: Lateral Femoral Condyle Medial Head: Medial Femoral Condyle	Posterior Surface of Calcaneus via Calcaneal Tendon	Tibial Nerve	Plantarflexes Ankle when Knee is Extended; Raises Heel During Walking; Flexes Leg at Knee
Soleus	Posterior Fibular Head, Superior ¼ of Posterior Fibula, and Soleal Line of Tibia	Posterior Surface of Calcaneus via Calcaneal Tendon		Plantarflexes Ankle
Plantaris	Supracondylar Line of Femur	Posterior Surface of Calcaneus via Calcaneal Tendon		Weakly Plantarflexes Ankle





E. Deep Posterior Compartment

a. There are four muscles in the deep posterior compartment and they all have a common innervation: **the tibial nerve.**
b. **Tarsal Tunnel:** Similar to the carpal tunnel, the tarsal tunnel contains the muscle tendons that pass from the deep posterior compartment of the leg to the foot. The tarsal tunnel is under the **flexor retinaculum** posterior to the **medial malleolus.**
   i. Contents of the tarsal tunnel include: **tendons of tibialis posterior, flexor digitorum longus and flexor hallucis longus**.
   ii. Also passing through the tunnel is the **posterior tibial artery** and the **tibial nerve.**
   iii. These contents of the tarsal tunnel can be remembered from ***anterior*** to ***posterior*** with the mnemonic:
      1. **Tom, Dick, And Not Harry** (Tibialis posterior, flexor Digitorum longus, posterior tibial Artery, tibial Nerve, and flexor Hallucis longus)
         a. **Note** that in the posterior leg, the order of the muscle bellies is different to the order of the tendons posterior to the medial malleolus. In the posterior leg from medial to lateral the order is flexor digitorum longus, tibialis posterior, flexor hallucis longus.

Clinical Note: Tarsal tunnel syndrome - the tibial nerve can become compressed in the tarsal tunnel by swelling of the synovial sheaths of the muscle tendons in the tunnel. The most common symptom is heel pain caused by compression of the tibial nerve by the flexor retinaculum.

Popliteus

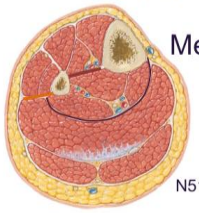






Origin:
Lateral Condyle of Femur and Lateral Meniscus

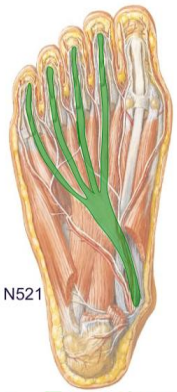
Insertion:
Posterior Surface of Tibia Superior to Soleal Line



- Function:
  - Weakly Flexes Knee
  - Unlocks Knee

Flexor Digitorum Longus

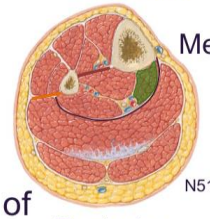






Origin:
Posterior Tibia Inferior to Soleal Line

Insertion:
Distal Phalanges of Lateral Four Digits

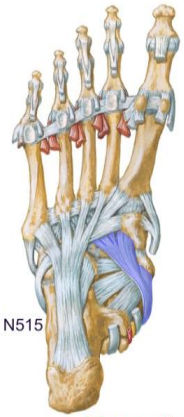


- Function:
  - Flexes Lateral Four Digits
  - Plantarflexes Ankle
  - Supports Longitudinal Arches of Foot

Tibialis Posterior

Myotome: L5

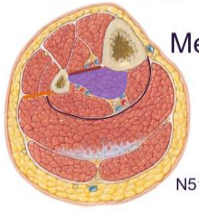






Origin:
Posterior Tibia, Fibula, and Interosseous Membrane

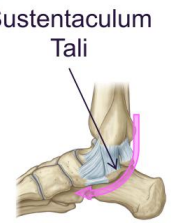
Insertion:
Primarily on Tuberosity of Navicular. Also on Cuneiforms, Cuboid, Calcaneus and 2nd – 4th MTs



- Function:
  - Plantarflexes Ankle
  - Inverts Foot
  - Supports Longitudinal Arches of Foot

Flexor Hallucis Longus

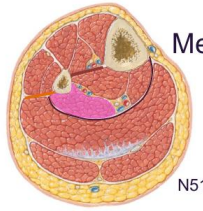






Origin:
Inferior ⅔ of Posterior Fibula; Inferior Interosseous Membrane

Insertion:
Base of Distal Phalanx of 1st Digit (Hallux)



- Function:
  - Flexion of 1st Digit
  - Weakly Plantarflexes Ankle
  - Supports Medial Longitudinal Arch of Foot

Muscle	Origin	Insertion	Innervation	Action
Popliteus	Lateral Condyle of Femur and Lateral Meniscus	Posterior Surface of Tibia Superior to Soleal Line	Tibial Nerve	Weakly Flexes Knee. Unlocks Knee by Laterally Rotating Femur on Fixed Tibia or Medially Rotating Tibia on Unplanted Limb.
Flexor Hallucis Longus (FHL)	Inferior 2/3 of Posterior Fibular; Inferior Interosseous Membrane	Base of Distal Phalanx of 1st Digit (Hallux)		Flexion of 1st Digit; Weakly Plantarflexes Ankle; Supports Medial Longitudinal Arch of Foot.
Flexor Digitorum Longus (FDL)	Posterior Tibia Inferior to Soleal Line.	Distal Phalanges of Lateral Four Digits.		Flexes Lateral Four Digits; Plantarflexes Ankle; Supports Longitudinal Arches of Foot.
Tibialis Posterior (TP)	Posterior Tibia Inferior to Soleal Line; Posterior Fibula; Interosseous Membrane.	Primarily on Tuberosity of Navicular. Also on Cuneiforms, Cuboid, Calcaneus, and 2nd-4th Metatarsals		Plantarflexes Ankle and Inverts Foot.

Knowledge Check: Which Muscles are Working?



Standing Calf Raise

gastrocnemius

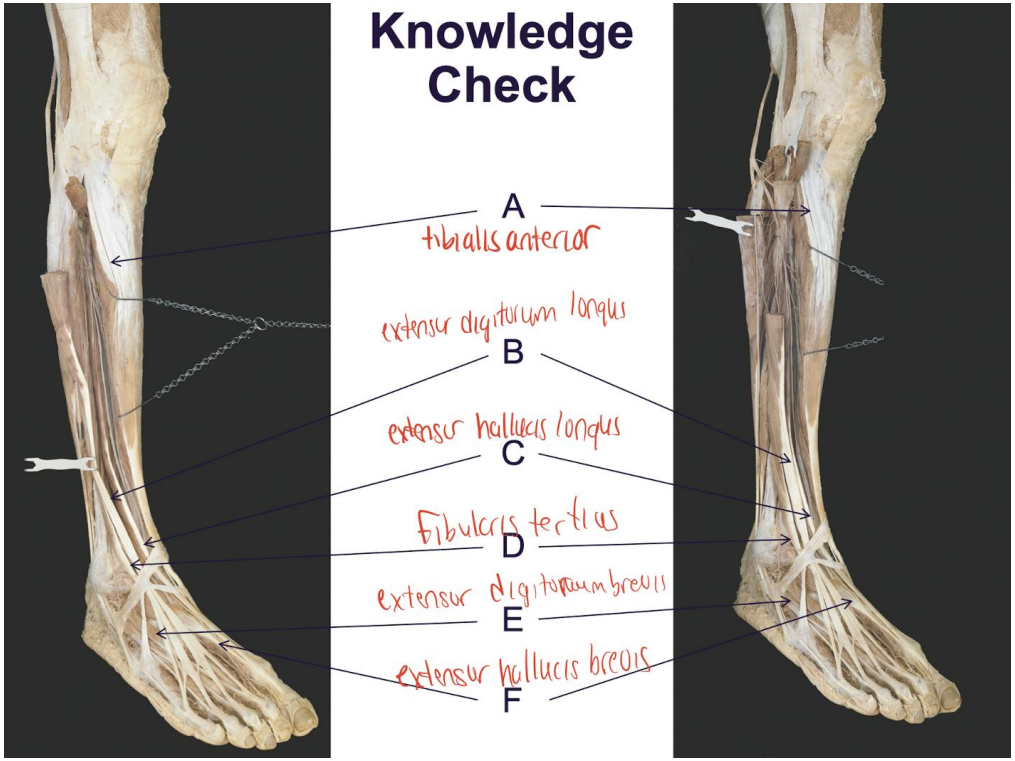
soleus fxning either way



Seated Calf Raise

Soleus

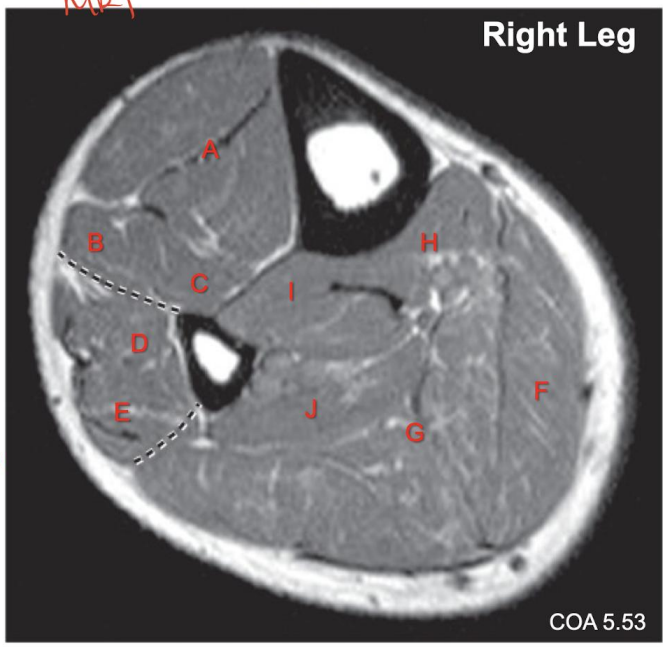
Knowledge Check



Knowledge Check

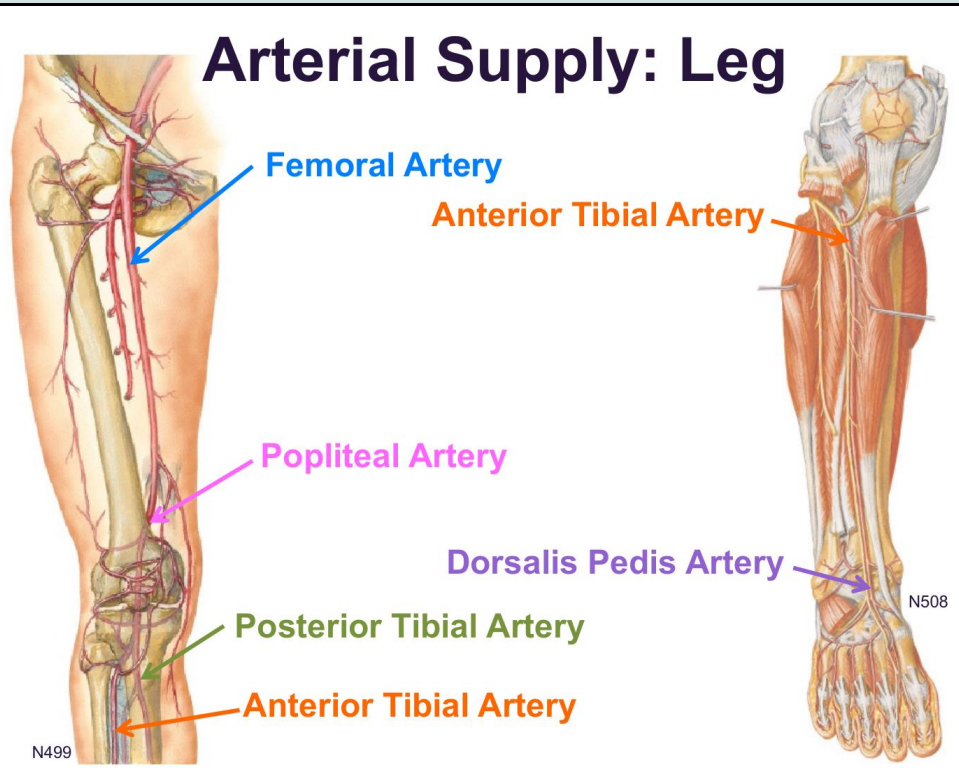
MRI

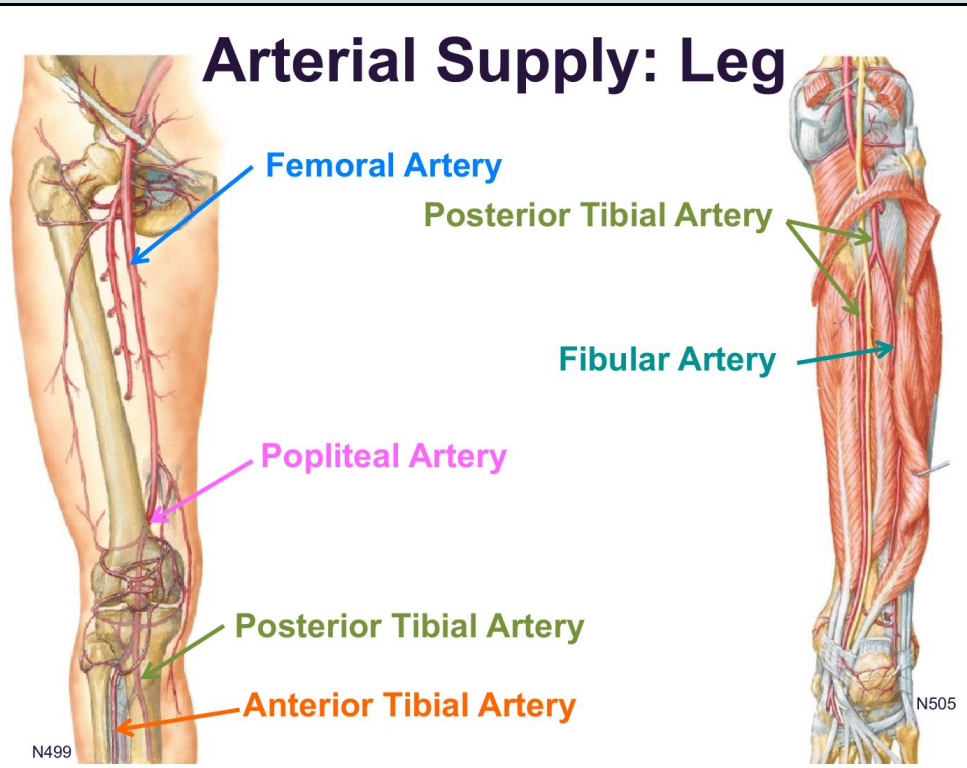
A) tibialis ant
B) EDL
C) EHL
D) FB
E) FL
G) Soleus
F) MG medial gastro
H) FLD
I) - tp
J) FHL



Neurovasculature

- Arterial Supply
- Venous Drainage





V. Arteries

a. **Review:** Aorta → Common Iliac Artery → External Iliac Artery → Femoral Artery → Popliteal Artery

b. At the inferior border of the popliteal fossa the **popliteal artery** becomes the **anterior and posterior tibial arteries.**

   i. **Anterior Tibial Artery:** the anterior tibial artery courses anteriorly through a gap in the superior part of the interosseous membrane before traveling inferiorly with the **deep fibular nerve** in the **anterior compartment.**

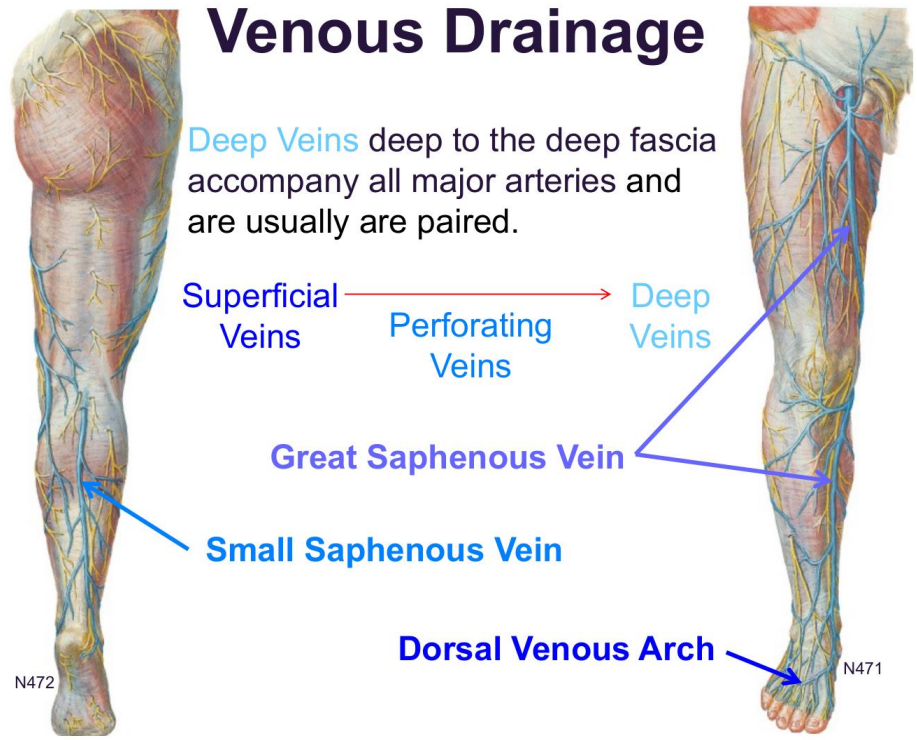
      1. **Dorsalis Pedis** – terminal branch of the anterior tibial artery starting between the malleoli. The dorsalis pedis courses on the dorsum of the foot giving off the lateral tarsal artery and **arcuate artery** before giving off the **deep plantar artery** and **1st dorsal metatarsal artery** as terminal branches.

         a. **Arcuate Artery** – a branch of dorsalis pedis. The arcuate artery comes off at the base of the 2nd metatarsal and courses laterally along the dorsal side of the metatarsals before anastomosing with the lateral tarsal artery. The arcuate artery gives rise to the 2nd, 3rd, and 4th dorsal metatarsal arteries.

   ii. **Posterior Tibial Artery:** arises with the anterior tibial artery. Courses within the deep posterior compartment with the **tibial nerve** and supplies the posterior compartment of the leg. The artery passes posterior to the medial malleolus within the **tarsal tunnel** to reach the foot.

      1. **Fibular Artery:** close to the origin of the **posterior tibial artery** it will give rise to its largest branch, the **fibular artery**.

         a. The fibular artery descends obliquely toward the fibula and passes inferiorly within **flexor hallucis longus.** The **fibular artery** gives muscular branches to the muscles of the posterior and lateral compartments.



VI. Veins

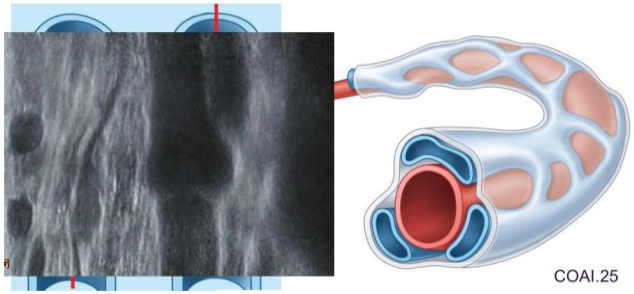
a. The lower limb has superficial and deep veins; the superficial veins are in the subcutaneous tissue, while the deep veins are deep to the deep fascia and accompany all major arteries.
b. Both superficial and deep veins have valves, which are more numerous in deep veins.

C. Superficial Veins:

1. **Great Saphenous Vein** – begins as the union of the **dorsal venous arch** and the dorsal vein of the great toe. The great saphenous vein courses anterior to the medial malleolus, up the medial side of the leg in the superficial fascia, behind the medial condyle, to the superior thigh, through the **saphenous opening** in the fascia lata, and into the **femoral vein.** As it ascends in the leg and thigh, the great saphenous vein receives numerous tributaries and communicates in several locations with the **small saphenous vein.**
2. **Small Saphenous Vein** – begins as the union of the **dorsal venous arch** and the dorsal vein of the little toe. Travels lateral to the calcaneal tendon in the superficial fascia before piercing the deep fascia and coursing through the two heads of the gastrocnemius muscle and draining into the **popliteal vein.**

Venous Return

Blood return to heart against gravity



COAI.25



COAI.26

Valves

- Allow blood to flow back to heart
- While preventing backflow

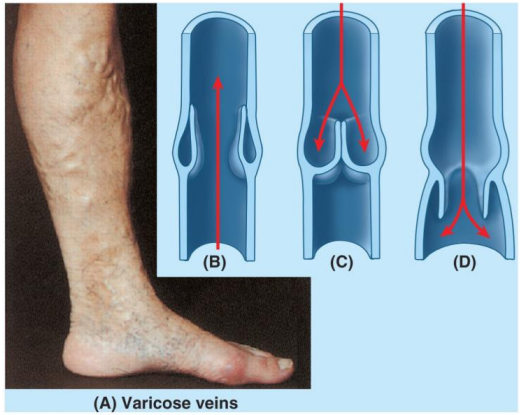
Arteriovenous Pump

- Double or multiple deep veins surround deep arteries
- Veins are compressed as the artery expands which aids in blood return to heart

Musculovenous Pump

- Outward expansion of contracting muscle bellies compresses veins
- "Milking" the blood toward the heart

Varicose Veins







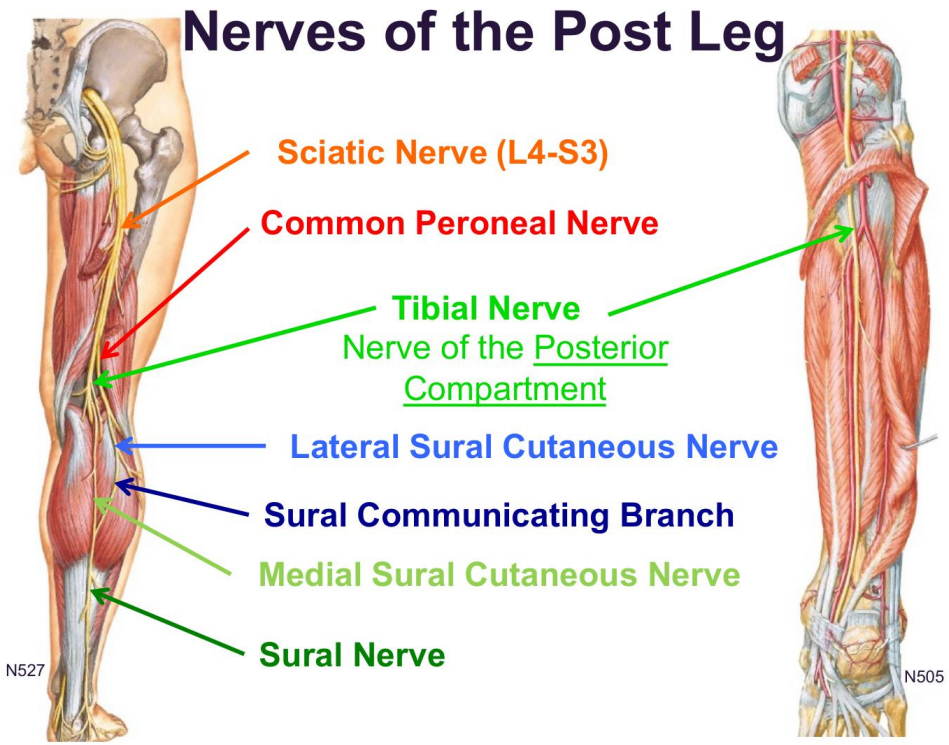
- The great saphenous vein and its tributaries become dilated so that the cusps of their valves do not close (D).
- As a result blood flows inferiorly producing **varicose veins**
- **Laser Treatment –** Damages vein causing scar tissue to form closing the vein. Van den Bos R, et al. (2009). Endovenous therapies of lower extremity varicosities: A meta-analysis. *Journal of Vascular Surgery*, 49(1): 230–239.

Venous return to the Heart.

1. **Venous Valves:** Most veins contain valves that assist blood return to the heart by preventing backflow of blood. The venous valves contain little cusps, when the cusps fill with blood they occlude the lumen of the vein preventing reflux of the blood distally making blood flow unidirectional.
2. **Arteriovenous Pump:** Although depicted as single vessels in illustrations for simplicity, veins tend to be double or multiple. The accompanying veins that travel with the deep arteries occupy a relatively unyielding fascial vascular sheath; thus, when the artery expands during contraction of the heart the veins are stretch and flattened, which aids in driving blood back toward the heart. This arrangement also serves as a countercurrent heat exchanger.
3. **3. Musculovenous Pump:** The deep accompanying veins lay deep to the deep fascia with the muscles of the thigh and leg. When the muscles contract their muscle bellies expand and increase girth, limited by the deep fascia the veins become compressed. The compression of the veins essentially "milks" the blood superiorly toward to the heart.

Perforating veins penetrate the deep fascia to connect the superficial veins to the deep veins. The perforating veins contain valves that allow blood to flow only from the superficial veins to the deep veins.

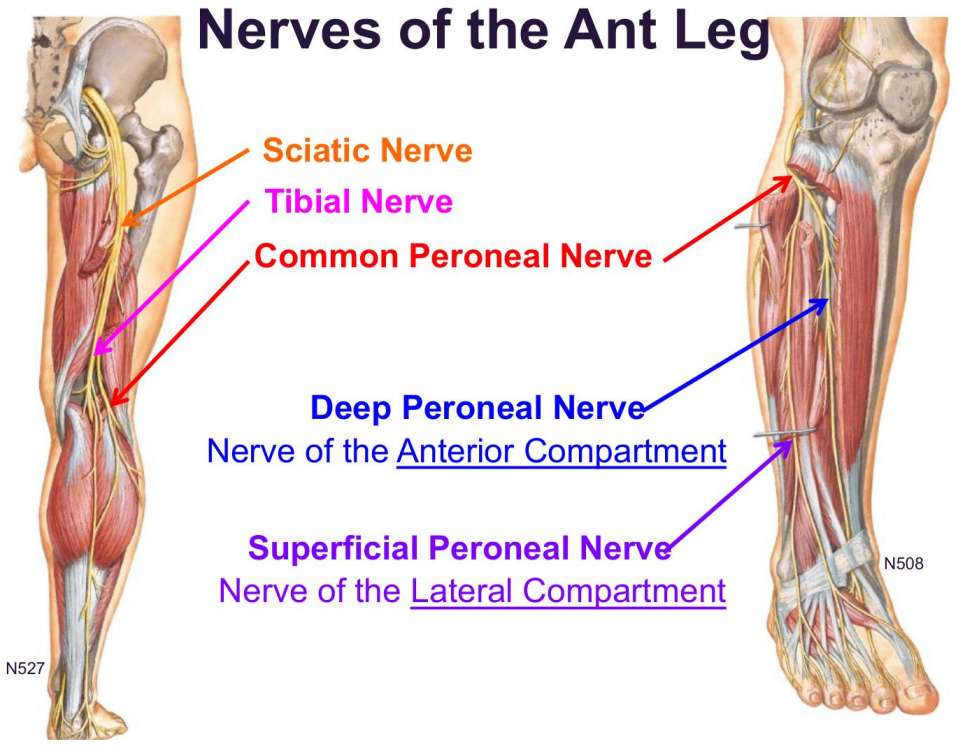
- **Clinical Note: Varicose Veins** – The great saphenous vein and its tributaries become dilated so that the cusps of their valves do not close, as a result blood flows distally in the veins, producing varicose veins. Varicose veins are common in the posteromedial parts of the lower limb and may cause discomfort. Laser treatment has become a safe and effective method of treatment of varicose veins. The heat from the laser damages the vein, which makes scar tissue form. This scar tissue closes the vein; the closed vein then loses its source of blood and dies. After a year or two, the vein is likely to disappear.



VII. Nerves

A. Motor Innervation:

1. **Sciatic Nerve (L4-S3):** travels through the posterior thigh. In actuality the sciatic nerve is two nerves, tibial and common fibular nerves, sharing a fascial sheath. The tibial and common fibular nerves separate from one another at the popliteal fossa. You will cover this material in detail in the Gluteal Region and Posterior Thigh and the Posterior Leg and Plantar Foot lectures.
    a. **Common Fibular Nerve** – continuation of sciatic nerve with tibial nerve. The common fibular nerve courses through the lateral popliteal fossa before wrapping around the neck of the fibula laterally to the anterior side where it gives off its two branches: the ***superficial fibular nerve*** and ***deep fibular nerve.***
    b. (continued) In order to compensate for footdrop and to prevent the toes from hitting the ground during normal gait, patients will adjust the swing phase of their gait by doing one or multiple of the following: raising their hip higher on the injured side (waddling gait), swinging the injured leg out laterally (swing-out gait), or raising the entire leg higher (steppage gait). Loss of dorsiflexion will also make landing on the heel of the foot difficult while walking, as a result, patients will swing their leg farther forward in order to have their heel hit. Lastly, loss of dorsiflexion of the foot is also loss of eccentric plantar flexion. Normal eccentric contraction of the dorsiflexors lets the anterior foot down slowly after the heel has struck; but without the muscles of the anterior compartment, the foot will not be let down in a controlled manner.
        i. **Superficial Fibular Nerve** – branch of the common fibular nerve and is the nerve of the ***lateral compartment***. The superficial fibular nerve travels in the lateral compartment where it innervates ***fibularis longus*** and ***fibularis brevis*** before piercing the crural fascia to become subcutaneous. See discussion of ***subcutaneous superficial fibular nerve*** below.
        ii. **Deep Fibular Nerve** – branch of the common fibular nerve and is the nerve of the ***anterior compartment***. The deep fibular nerve arises between the fibularis longus muscle and the neck of the fibula. After it enters the anterior compartment it accompanies the ***anterior tibial artery*** where it innervates ALL muscles of the anterior compartment. It exits the anterior compartment distally to enter the dorsum of the foot to innervate the muscles of the dorsal foot. Finally, the deep fibular nerve gives cutaneous innervation to the space between the big and 2nd toe.



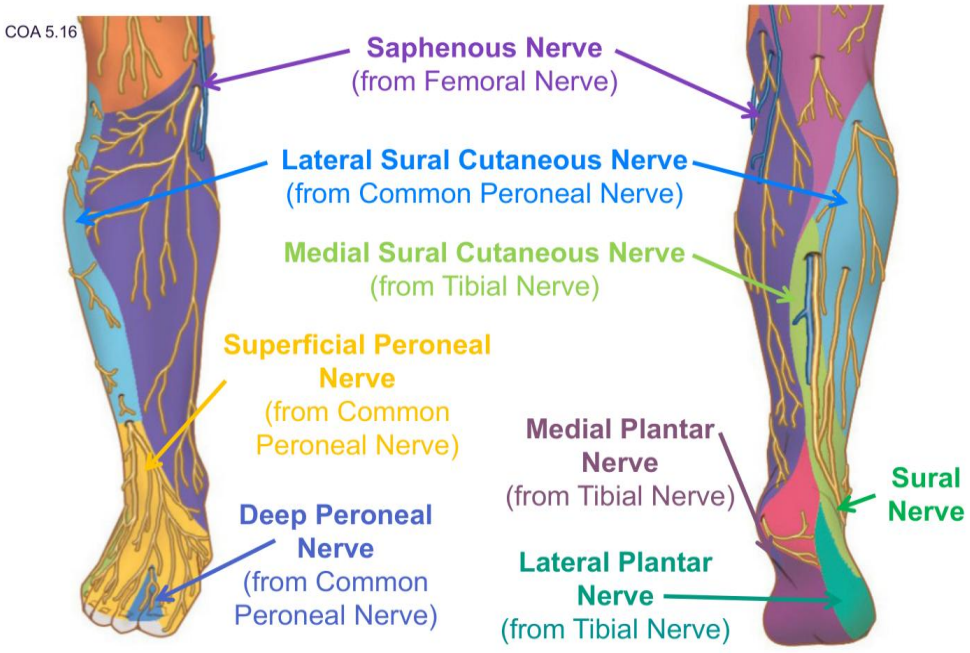
Clinical Note: Superficial Fibular Nerve Entrapment - Numerous ankle sprains can stretch the superficial fibular nerve to the point of injury.

- Such an injury can cause pain along the anterolateral leg and dorsum of the foot as well as paresthesia in these areas and may increase with activity.

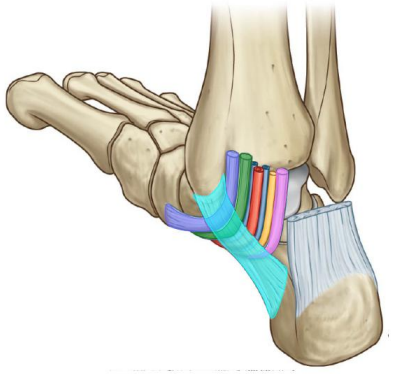
Clinical Note: Deep Fibular Nerve Entrapment - When the muscles of the anterior compartment are overused and inflamed, the deep fibular nerve can become compressed.

- Also, tight boots such as ski boots or soccer cleats can compress the deep fibular nerve where it passes under the inferior extensor retinaculum. In either case, the compression can cause pain on the dorsum of the foot that radiates to the space between the big and 2nd toe.

Sensory Nerve Supply



Tom, Dick, And Not Harry



Tarsal Tunnel

Anterior ⟷ Posterior

Tibialis Posterior
Flexor Digitorum Longus
Posterior Tibial Artery
Tibial Nerve
Flexor Hallucis Longus

B. Sensory Innervation

1. **Saphenous Nerve** – continuation of the femoral nerve in the anterior leg; supplies sensation to the medial side of the leg and foot.
2. **Lateral Sural Cutaneous Nerve** - a branch of the common fibular nerve; supplies sensation to the posterolateral portion of the leg.
3. **Superficial Fibular Nerve** – a branch of the common fibular nerve; supplies sensation to the anterolateral leg and dorsum of the foot (not including the space between the big and second toe).
4. **Deep Fibular Nerve** – a branch of the common fibular nerve; supplies sensation to the space between the big and second toe.
5. **Lateral Dorsal Cutaneous Nerve of the Foot** – termination of sural nerve; supplies sensation to the lateral foot